

RESEARCH ARTICLE

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Key Points:

- We estimate the effective plant rooting depth (Z_r) using a carbon cost-benefit model across global terrestrial ecosystems
- Both mean climate conditions and climate seasonality are essential in determining Z_r
- Z_r estimated herein is more hydrologically effective than existing global Z_r data sets

Supporting Information:

- Supporting Information S1

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Global estimation of effective plant rooting depth: Implications for hydrological modeling

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Abstract Plant rooting depth (Z_r) is a key parameter in hydrological and biogeochemical models, yet the global spatial distribution of Z_r is largely unknown due to the difficulties in its direct measurement. Additionally, Z_r observations are usually only representative of a single plant or several plants, which can differ greatly from the effective Z_r over a modeling unit (e.g., catchment or grid-box). Here, we provide a global parameterization of an analytical Z_r model that balances the marginal carbon cost and benefit of deeper roots, and produce a climatological (i.e., 1982–2010 average) global Z_r map. To test the Z_r estimates, we apply the estimated Z_r in a highly transparent hydrological model (i.e., the Budyko-Choudhury-Porporato (BCP) model) to estimate mean annual actual evapotranspiration (E) across the globe. We then compare the estimated E with both water balance-based E observations at 32 major catchments and satellite grid-box retrievals across the globe. Our results show that the BCP model, when implemented with Z_r estimated herein, optimally reproduced the spatial pattern of E at both scales (i.e., $R^2 = 0.94$, RMSD = 74 mm yr^{−1} for catchments, and $R^2 = 0.90$, RMSD = 125 mm yr^{−1} for grid-boxes) and provides improved model outputs when compared to BCP model results from two already existing global Z_r data sets. These results suggest that our Z_r estimates can be effectively used in state-of-the-art hydrological models, and potentially biogeochemical models, where the determination of Z_r currently largely relies on biome type-based look-up tables.

1. Introduction

In terrestrial ecosystems, the plant rooting depth (Z_r) primarily determines the active soil zone that has potential to return water back into the atmosphere via plant transpiration. It also profoundly affects the exchange of energy and carbon between the land surface and the atmosphere, due to the close coupling among the terrestrial water, energy and carbon cycles [e.g., Guswa, 2010; Ichii *et al.*, 2007; Kleidon and Heimann, 1998; Law *et al.*, 2002; Yang *et al.*, 2013, 2015b]. Besides its critical role in regulating the land surface-atmosphere interactions, a correct parametrization of Z_r is essential in many hydrological and biogeochemical models to successfully simulate subsurface processes, such as deep drainage and runoff generation [Milly, 1994].

Despite its importance in the control of terrestrial hydrological and biogeochemical processes, unfortunately the global distribution of Z_r and its relationships with climate, soil, and plant physiological characteristics remains largely unknown. This is primarily due to the difficulties in the direct quantification of Z_r using field approaches, especially for deeper-rooted plants. A few studies have attempted to compile global Z_r observations from the literature [e.g., Canadell *et al.*, 1996; Jackson *et al.*, 1996; Schenk and Jackson, 2002a, 2002b]. Among these reviews, Schenk and Jackson [2002a] probably represents the most comprehensive collection of Z_r observations to-date, where 475 root profiles (with less than 4% sampled to depths greater than 3 m) from 209 locations around the world were summarized. They later extrapolated these observations to produce a global map of Z_r that contains upper 95% of root biomass based on the statistical relationships between biome-averaged Z_r and climate and soil properties [Schenk and Jackson, 2009]. However, this data-driven method is inevitably subject to uncertainties caused by: (i) data scarcity; (ii) the scale mismatch between input variables (i.e., the vastly different scales at which Z_r was measured and the scale at which climate and soil variables were represented in the modeling); and (iii) the reliability of the constructed statistical relationships in predicting Z_r .

In light of the lack of Z_r observations and the importance of Z_r in many research fields, a number of models have been proposed to estimate Z_r directly, or indirectly, based on various scientific hypotheses (reviewed in Wang-Erlandsson *et al.* [2016]). For direct modeling, the optimality theory-based approaches have been

mostly used [Collins and Bras, 2007; Eagleson, 2004; Hildebrandt and Eltahir, 2007; Kleidon, 2004; Kleidon and Heimann, 1998; Laio et al., 2006; van Wijk and Bouten, 2001]. The scientific basis of the optimality models is that, for a given climate and soil environment, the optimal Z_r allows plants to minimize their water stress over a long-time period (i.e., climatologically) and to maximize transpiration and/or productivity. These models have been widely used to examine the role of climatic, edaphic, and physiological factors on Z_r ; nevertheless, they have not yet been extensively tested nor broadly applied across different environments [Collins and Bras, 2007; Schenk, 2008; Wang-Erlandsson et al. 2016]. One of the few existing global applications of the optimality models of Z_r is presented in Kleidon and Heimann [1998] and later updated in Kleidon [2004], where a light use efficiency model of vegetation photosynthesis is coupled with a simple soil hydrological model to obtain a global map of the optimal Z_r that results in the highest vegetation net primary production (or more fundamentally, the maximum vegetation transpiration) over a 10 year period. One key limitation of this model lies in the assumption that vegetation fully covers the land surface over the entire globe. Consequently, when maximizing transpiration, it ignores the nonbiological water loss through soil evaporation, which can account for a large proportion of total actual evapotranspiration (E) in sparsely vegetated areas [Yang et al., 2015a; Yang and Shang, 2013; Long and Singh, 2012; Zhang et al., 2016]. In other words, Z_r estimates from Kleidon's [2004] model represent rooting depth per vegetated area rather than rooting depth per ground area which is more hydrologically and/or ecologically meaningful. In the context of hydrological and/or biogeochemical modeling, we are more interested in the effective rooting depth over an analysis unit (e.g., a catchment or grid-box) rather than the actual rooting depth of a single plant. Therefore, although Kleidon's [2004] model attempts to maximize the net carbon gain, it lacks the actual carbon constraint on Z_r , which is expressed via vegetation fraction and density. This issue similarly exists in other optimality models [e.g., Laio et al., 2006] and in data-driven models [e.g., Schenk and Jackson, 2009]. Theoretically, such a treatment would overestimate the effective Z_r for a specific location and therefore overestimate E and underestimate runoff for a given rainfall in hydrological modeling.

Indirect estimation of Z_r involves methods such as inverse modeling and model calibration [e.g., Campos et al., 2016; Kleidon, 2004; Winsemius et al., 2009; Ichii et al., 2009]. These two methods are essentially very similar, as they both require the target value of the model output (e.g., observations or estimates from other models) as prior knowledge and then find the optimal Z_r value that permits the model output to best match that target value. As a result, the inverse modeling and model calibration approaches are also subject to similar issues. The accuracy of Z_r estimates from inverse modeling and/or model calibration is largely contingent on the quality of the target variables and the model sensitivity to variations in Z_r . Moreover, when calibrating hydrological models, it is usually only applicable at the catchment scale and other difficulties such as parameter equifinality [Beven, 2006] bring additional uncertainties into the final Z_r estimates.

By considering the carbon budget associated with root growth, Guswa [2008] developed an analytical model to directly estimate Z_r that balances the carbon gain and cost of any additional roots. In comparison to other direct modeling approaches, the analytical nature of Guswa's model enhances insight by making transparent the dependencies among variables. In addition, the model explicitly incorporates the carbon constraint whilst considering the actual water stress (rather than to minimize water stress) on vegetation transpiration through the coupling of a dynamic soil moisture model [Milly, 1993; Rodriguez-Iturbe et al., 1999]. In this regard, Z_r from Guswa [2008] is expected to be more hydrologically and ecologically meaningful and further efforts to apply and test the model across the global range of bioclimates is warranted.

The objectives of this study are to: (i) provide a global parameterization of Guswa's [2008] model to produce a global Z_r map and examine the climatic controls on Z_r ; (ii) compare Z_r estimated herein with two other existing global Z_r data sets (i.e., (a) Z_r estimated by an optimality model [Kleidon, 2004]; and (b) up-scaled from in situ measurements [Schenk and Jackson, 2009]); and (iii) examine how different Z_r parameterizations affect hydrological simulation across global terrestrial ecosystems. To test the Z_r estimates and examine the effect of Z_r parameterizations in hydrological modeling, we ingest the estimated Z_r into a hydrological model and compare the model outputs with in situ observations and satellite-based modeling outputs of actual evapotranspiration. Validation of hydrological model outputs provides an indirect, yet effective, test of the hydrological effectiveness of Z_r estimates. The adopted hydrological model should be parsimonious and transparent while explicitly incorporating Z_r to minimize any potential effects of interactions between Z_r and other model parameters on the model outputs (see Method section for details).

2. Material and Methods

2.1. Estimation of Plant Rooting Depth Using *Guswa's* Carbon Cost-Benefit Model

Rooting depth was predicted using a carbon cost-benefit model developed in *Guswa* [2008]. The fundamental idea of the *Guswa's* model is that deeper roots permit plants to access more soil moisture, which allows plants to avoid water stress for longer times during meteorologically dry periods and therefore achieve greater gross carbon uptake. On the other hand, the construction and maintenance of additional roots consume carbon. As a result, there must be an optimal rooting depth at which the marginal carbon benefit associated with any additional roots balances the marginal carbon cost of those roots.

Following *Guswa* [2008], the average carbon cost and benefit of plant roots can be written, respectively, as:

$$C(Z_r) = \int_0^{Z_r} \frac{\gamma_r RLD}{SRL} dZ_r \quad (1)$$

$$B(Z_r) = WUE \times f_{GS} \times (E_{T,Z_r}) \quad (2)$$

where Z_r is plant rooting depth (cm). C and B represents carbon cost and benefit of roots, respectively ($\text{g C cm}^{-2} \text{ d}^{-1}$). γ_r is the root respiration rate ($\text{g C g}^{-1} \text{ roots d}^{-1}$), RLD is the root length density (cm roots cm^{-3} soil), and SRL is the specific root length ($\text{cm roots g}^{-1} \text{ roots}$). WUE is the photosynthetic water use efficiency ($\text{g C cm}^{-3} \text{ H}_2\text{O}$), f_{GS} represents the fraction of growing season within a year, and (E_{T,Z_r}) is the mean annual transpiration rate for a given rooting depth (mm d^{-1}).

A simple dynamic soil moisture model of *Milly* [1993] was then incorporated to estimate (E_{T,Z_r}) as a function of potential transpiration rate (PT), soil moisture availability, and Z_r . In *Milly* [1993], rainfall is considered as a stochastic process and the climate seasonality is parametrized through the combination of total rainfall depth (P) and the mean rainfall depth per event (α). Because statistically, a smaller ratio of P over α indicates a longer mean time between rainfall events (the ratio of P over α represents the number of events), which may indicate a longer dry season (the dry season was identified when mean monthly potential evapotranspiration is larger than that of precipitation [*Kleidon and Heimann*, 1998], see supporting information Figure S1). A full description of how E_T was determined is given in *Guswa* [2008] and *Milly* [1993]. It is worth noting that PT is the atmospheric water requirement from the vegetation component only and is different from potential evapotranspiration (E_p) that has been more conventionally used in hydrology and climate studies. In *Guswa* [2008], PT was estimated by subtracting soil evaporation and canopy interception from E_p , which requires the maximum evaporation depth per rainfall event as prior knowledge (i.e., equation (7) in *Guswa* [2008]). However, this parameter is usually not available in most hydrometeorological observations. Here, we calculated PT using the hybrid dual-source E_p model of *Guan and Wilson* [2009], which is a physically based model that partitions E_p into PT , potential evaporation from soil and canopy interception, and requires only readily-available near-surface climate (i.e., net all-wave radiation, air temperature, vapor pressure, and wind speed) and vegetation (i.e., satellite-derived vegetation indices) data as inputs.

The resultant equation for estimating Z_r is [see *Guswa*, 2008 for full details]:

$$Z_r = \frac{\alpha}{\theta(1-W)} \ln(X) \quad (3)$$

$$X = \begin{cases} W \left[1 + \frac{\theta(1-W)^2}{\alpha} - \sqrt{\frac{\theta(1-W)^2}{\alpha} + \left(\frac{\theta(1-W)^2}{2A}\right)^2} \right] & \text{if } W > 1 \\ W \left[1 + \frac{\theta(1-W)^2}{\alpha} + \sqrt{\frac{\theta(1-W)^2}{\alpha} + \left(\frac{\theta(1-W)^2}{2A}\right)^2} \right] & \text{if } W < 1 \end{cases} \quad (4)$$

$$A = \frac{\gamma_r \times RLD}{SRL \times WUE} \times \frac{1}{PT \times f_{GS}} \quad (5)$$

where W is the unitless ratio of mean annual P over PT , and θ is the plant-available water content ($\text{cm}^3 \text{ cm}^{-3}$), which is the product of soil porosity (ϵ , $\text{cm}^3 \text{ cm}^{-3}$) and the difference between soil field capacity and wilting point (i.e., $\theta_{fc} - \theta_w$).

2.2. Determining Model Parameters

Seven parameters (i.e., γ_r , RLD , SRL , WUE , f_{GS} , α , and θ) need to be determined prior to applying the model to estimate Z_r . Root respiration rate (γ_r) was quantified using the Q_{10} equation, which is a function of air temperature:

$$\gamma_r = \gamma_{r_{20}} \times Q_{10}^{\left(\frac{T-20}{10}\right)} \quad (6)$$

where $\gamma_{r_{20}}$ is the root respiration rate at 20°C, T is air temperature (°C), and Q_{10} is a temperature coefficient that represents the factor by which the root respiration rate changes for every 10°C rise in the air temperature [Lloyd and Taylor, 1994]. The value of Q_{10} for roots is fixed at 2.0 following White *et al.* [2000] and that of $\gamma_{r_{20}}$ is determined from a look-up table for each vegetation type [Running and Zhao, 2015]. Because there are currently no global data sets of RLD and SRL available, we determined RLD to be 0.1 cm roots cm⁻³ soil and SRL to be 1500 cm roots g⁻¹, which, respectively, represent the medium value of these two parameters reported in the literature [Caldwell, 1994; Eissenstat, 1997; Fitter and Hay, 2002; Pregitzer *et al.*, 2002; Scholes and Walker, 1993]. The spatial pattern of WUE was quantified using annual gross primary production (GPP) and E_T estimates from eight process-based ecosystem models [Huntzinger *et al.* [2013]; also see Data for details and supporting information Figure S2). The ensemble mean of model outputs was used as it effectively reduces the uncertainty in the estimates by individual models [Anav *et al.*, 2015].

The growing season length was identified from the satellite-observed leaf area index (LAI) data set. The Savitzky-Golay filter is first used to smooth the LAI data to minimize various atmospheric effects [Chen *et al.*, 2004]. Using the smoothed LAI time series, the beginning of the growing season was defined as when LAI is larger than 0.1 and has increased by 15% of the amplitude of the growing season, and the end of the growing season was determined to be the time when LAI value is greater than 0.1 and has decreased by 15% of the amplitude of the growing season [Zhu *et al.*, 2016]. Then, the initially determined growing season from the above procedures was further refined by excluding the ground-freeze period determined from a global Freeze/Thaw data set [Zhu *et al.*, 2013, 2016; Kim *et al.*, 2011] (see supporting information Figure S3). Mean rainfall depth (α) was quantified based on a global 3 hourly precipitation data set [Weedon *et al.*, 2014], and θ was determined from soil texture data and the U.S. Department of Agriculture (USDA) soil classification [Saxton and Rawls, 2006]. A summary of parameters required by Guswa's [2008] Z_r model including their quantification and value range is given in Table 1.

2.3. Implications of Z_r in Hydrological Modeling: The Budyko-Choudhury-Porporato (BCP) Model

We adopted the Budyko-Choudhury-Porporato (BCP) model [Donohue *et al.*, 2012] to test the estimated Z_r and to examine the effect of different Z_r parameterizations on modeled E . The BCP model combines Choudhury's expression of the Budyko curve [Choudhury, 1999] with a stochastic dynamic soil moisture model of Porporato *et al.* [2004] to quantify the hydrological partitioning at steady state conditions (i.e., mean annual averages). Choudhury's [1999] formulation of the Budyko curve is written as:

$$E = \frac{PE_p}{(P^n + E_p^n)^{1/n}} \quad (7)$$

where P is the precipitation depth (mm). n is a model parameter that encodes land surface properties and climate variabilities, and modifies the partitioning of P between E and runoff (Q).

Table 1. Summary of Parameters Needed in the Guswa's Z_r Model

Parameter (Unit)	Method of Determination	Value/Value Range	References
$\gamma_{r_{20}}$ (g C g ⁻¹ roots d ⁻¹)	Look-up table	5.19 ~ 8.19 ($\times 10^{-3}$)	Running and Zhao [2015]
RLD (cm roots cm ⁻³ soil)	Medium of reported values	0.1	Caldwell [1994] Eissenstat [1997]
SRL (cm roots g ⁻¹ roots)	Medium of reported values	1500	Fitter and Hay [2002] Pregitzer <i>et al.</i> [2002] Scholes and Walker [1993]
WUE (g C cm ⁻³ H ₂ O)	Based on ecosystem model outputs	0–1.6	Huntzinger <i>et al.</i> [2013]
f_{GS} (Dimensionless)	Based on satellite LAI series and refined by excluding freeze period	0–1	Zhu <i>et al.</i> [2016]
α (mm)	Based on 3 hourly precipitation	0.2–61.7	Weedon <i>et al.</i> [2014]
θ (cm ³ cm ⁻³)	Based on soil properties	0–0.52	Nachtergaele <i>et al.</i> [2009]

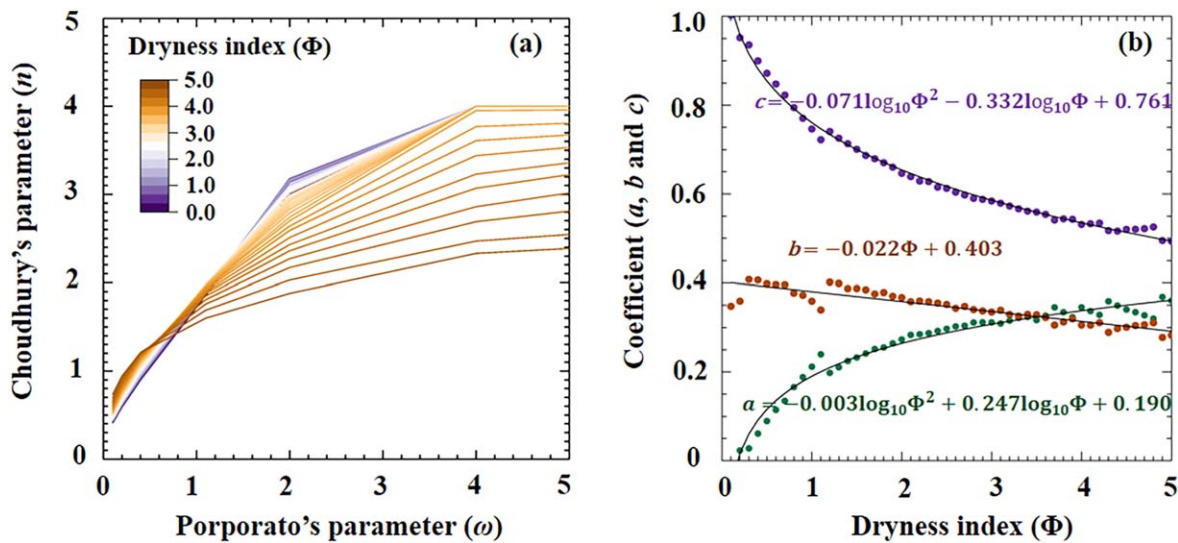


Figure 1. Relationship between Choudhury's parameter n and Porporato's parameter ω . (a) The relationship between n and ω , as a function of climate dryness index (Φ); and (b) illustrates the relationship between regression coefficients (i.e., a , b , and c in equation (10)) and Φ .

The probabilistic steady state solution of Porporato's stochastic dynamic soil moisture model (i.e., equations (1–5) in Porporato *et al.* [2004]) shares a similar form with the Budyko curve. Porporato's parameter (here we use ω instead of γ per Porporato *et al.* [2004] to distinguish it from the root respiration parameter used in Guswa's Z_r model) is a function of Z_r , α and θ (i.e., $\omega = Z_r\theta/\alpha$) and exhibits a close relationship with Choudhury's parameter n . Donohue *et al.* [2012] deduced the relationship between n and ω as:

$$n = 0.21\omega + 0.6 = 0.21 \frac{Z_r\theta}{\alpha} + 0.6 \quad (8)$$

However, as described in Donohue *et al.* [2012], equation (9) is only strictly valid when the dryness index (i.e., $\Phi = E_p/P$) equals to 1. Here, through numerically solving equation (8) with data obtained from Porporato *et al.* [2004], we rededuced the relationship between n and ω by considering this relationship as a function of Φ . The resultant functions for estimating n from ω are given as (also see Figure 1):

$$n = a\log_{10}\omega^2 + b\log_{10}\omega + c \quad (9)$$

where

$$\begin{cases} a = -0.003\log_{10}\Phi^2 + 0.247\log_{10}\Phi + 0.190 \\ b = -0.022\Phi + 0.403 \\ c = -0.071\log_{10}\Phi^2 - 0.332\log_{10}\Phi + 0.761 \end{cases} \quad (10)$$

Equations (7), (9), and (10) constitute a modification of the BCP model of Donohue *et al.* [2012], and is used in this study to estimate mean annual E over 1982–2010 with three different parameterizations of rooting depths being: (i) Z_r estimated herein using Guswa's carbon cost-benefit model; (ii) Z_r estimated from an optimality model of Kleidon [2004]; and (iii) Z_r estimated by upscaling in situ measurements [Schenk and Jackson, 2009].

2.4. Data

Global monthly half-degree near-surface meteorological data over 1982–2010, including air temperature, atmospheric vapor pressure, and wind speed, were obtained from the Climate Research Unit (CRU) TS3.2 [Harris *et al.*, 2014]. Net all-wave radiation was acquired from the National Centers for Environmental Prediction (NCEP) reanalysis output (<http://dods.extra.cea.fr/data/p529viov/cruncep>). Global 3 hourly half-degree precipitation data were obtained from the WFDEI meteorological forcing data set [Weedon *et al.*, 2014]. These fine temporal resolution data (e.g., monthly or 3 hourly) were then aggregated to obtain mean annual climate forcing for the modeling. Land cover classification, including the evergreen needle-leaf forest

(ENF), evergreen broadleaf forest (EBF), deciduous needle-leaf forest (DNF), deciduous broadleaf forest (DBF), mixed forest (MF), shrubland, savanna, grassland, cropland, and desert vegetation, was acquired from the Moderate Resolution Imaging Spectroradiometer (MODIS) land use map (MOD12Q1) for the year of 2001 available from the NASA data center (<http://reverb.echo.nasa.gov/>; Friedl *et al.* [2010]). Monthly vegetation indices, including LAI and the fraction of absorbed photosynthetic active radiation (f_{PAR}), were derived from Zhu *et al.* [2013] based on AVHRR GIMMS-3g NDVI data [Pinzon and Tucker, 2014]. This satellite LAI product has been extensively evaluated against field measurements, validated as a research-quality data set and used in numerous previous studies to understand trends in the terrestrial carbon and water exchanges [Pfeifer *et al.*, 2014; Yang *et al.*, 2016; Zhu *et al.*, 2013, 2016]. Soil texture data at 30'' spatial resolution was acquired from the Harmonized World Soil Database (HWSD) [Nachtergaele *et al.*, 2009]. To obtain a spatial pattern of WUE , global monthly GPP and E_T estimates over 1982–2010 at 0.5° spatial resolution were obtained from eight ecosystem models from MsTMIP (Multi-scale Synthesis and Terrestrial Model Intercomparison Project; Huntzinger *et al.* [2013]), including: (i) CLM [Mao *et al.*, 2012]; (ii) CLM4-VIC [Li *et al.*, 2011]; (iii) ISAM [Jain *et al.*, 1996]; (iv) TRIPLEX [Peng *et al.*, 2002]; (v) LPJ-wsl [Sitch *et al.*, 2003]; (vi) ORCHIDEE-LSCE [Krinner *et al.*, 2005]; (vii) SiBCASA [Schaefer *et al.*, 2008]; and (viii) VISIT [Ito, 2010]. The rooting depth data of Kleidon [2004] and Schenk and Jackson [2009] (referred to as $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ hereafter) were both obtained from the Oak Ridge National Laboratory Distributed Active Archive Center (<http://daac.ornl.gov>). As needed, the data were resampled to a spatial resolution of 0.5° × 0.5° using either the majority (for land cover data) or the nearest neighbor (for all other data) sampling approaches.

Two global E data sets were used to validate the estimated E from the BCP model. First, observed E (calculated from catchment water balance) over 32 major river basins across the globe from 1984 to 2006 were obtained from the global water cycle assessment study of Pan *et al.* [2012] (see supporting information Figure S4 for the spatial distribution of 32 major catchments). Second, global monthly half-degree remote sensing-based E estimates from the Penman-Monteith-Leuning (PML) model over 1982–2010 were provided by Zhang *et al.* [2016], where E estimates are well-validated against observations at 95 flux-towers ($R^2 = 0.87$, bias = −1.4%) and water balance calculations at 643 catchments ($R^2 = 0.77$, bias = −6.0%) across the globe.

3. Results

3.1. Global Distribution of Z_r and the Relationship With Climate

The spatial distribution of Z_r as estimated from the *Guswa's* carbon cost-benefit model is shown in Figure 2, with a global mean rooting depth of 0.678 m. Deeper roots (i.e., typically > 1.5 m) are generally observed in tropical and subtropical regions, including the Amazon and Congo basins and southeast Asia. In contrast, in more arid and/or cold regions, the Z_r values are usually less than 0.4 m, suggesting a shallower

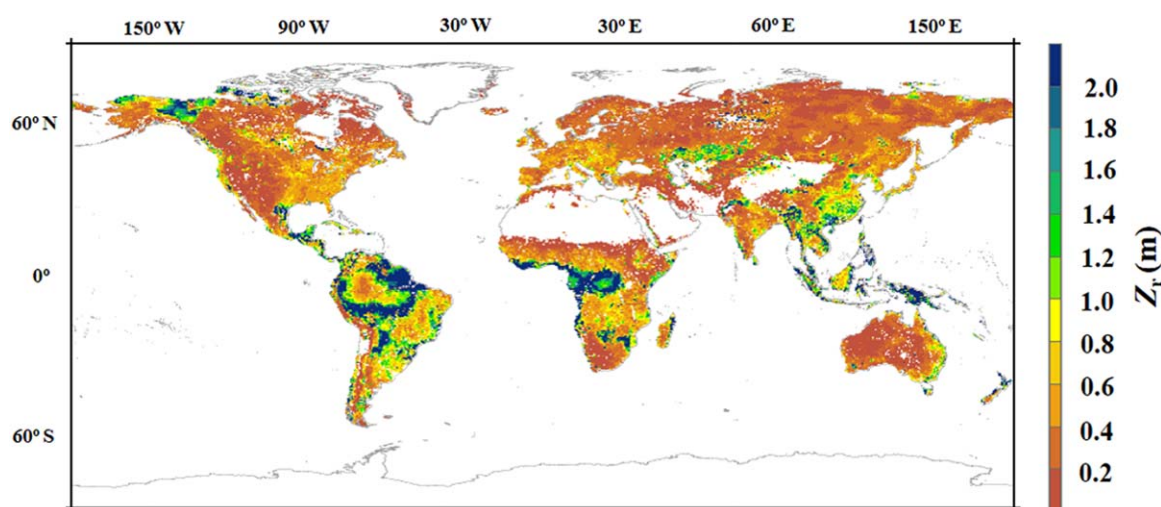


Figure 2. Global spatial pattern of effective plant rooting depth (Z_r) estimated from the *Guswa's* carbon cost-benefit model.

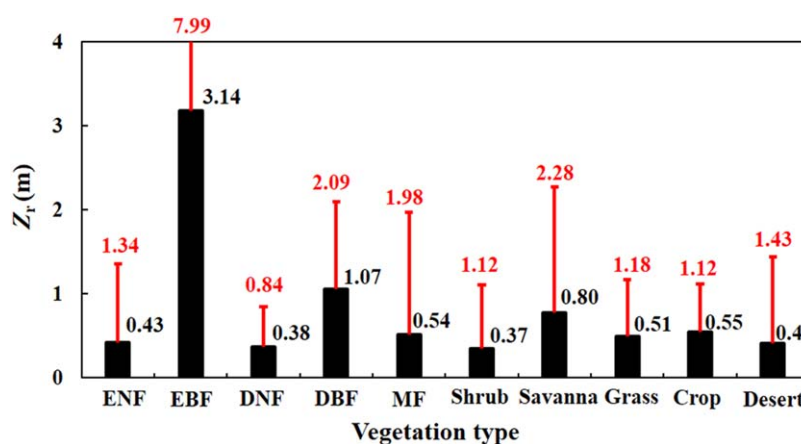


Figure 3. Averaged effective plant rooting depth (Z_r) for each vegetation type. Error bars indicate one standard deviation. Values for averaged Z_r , plus one standard deviation, for each vegetation type are given in black and red, respectively.

hydrologically active soil layer (or a smaller storage capacity). At the biome level, the deepest Z_r is observed in EBFs, which generally occur in tropical and subtropical regions (with a high Z_r spatial heterogeneity), followed by DBF and savanna (Figure 3). For the remaining seven vegetation types, their mean Z_r are all less than 0.6 m. The shallowest mean Z_r was found in shrubland (i.e., 0.35 m), which also encompasses most of the tundra ecosystems in the boreal region according to the MODIS land cover classification.

It is worthwhile highlighting again that Z_r estimated herein represents the effective rooting depth (accounting for both the vertical rooting depth and the spacing between plants) over an analyze unit (i.e., one pixel with an area of $0.5^\circ \times 0.5^\circ$) rather than the actual rooting depth of an individual plant. As a result, although quite deep roots in an order of 10 m have been often observed in arid climate zones [Canadell *et al.*, 1996; Schenk and Jackson, 2002b], the effective rooting depth in those areas are very small due to low vegetation coverage in these bioclimatic regions, indicating a high carbon constraint on Z_r .

Figure 4 depicts the relationship between Z_r and climate factors over the entire globe. Across the full range of dryness index (Φ , also known as the aridity index), Z_r shows an abrupt increase from extreme wet regions to the maximum value where Φ is around 0.8, and then decreases rapidly until $\Phi \sim 1.5$. Afterwards (i.e., $\Phi > 1.5$), even though the climatic dryness increases, the value of Z_r remains fairly constant (Figure 3a). This result is consistent with the inverse modeling of Z_r at 431 catchments [Gentine *et al.*, 2012] and Z_r observations along the Kalahari rainfall gradient [Bhattachan *et al.*, 2012]. For extreme humid regions where water is generally abundant throughout the year, plants are usually in a water-optimal condition and thus do not need to allocate more carbon to construct deeper roots but rather invest more carbon into the above-ground sector (e.g., more leaves) to maximize their light absorption [Field *et al.*, 1992]. On the other hand, in more arid zones, the shallower Z_r is more a reflection of the carbon constraint (i.e., vegetation cover and density) and in case of nonperennial species (e.g., grass), their assimilated carbon tends to be more invested into rapid growth and seeding instead of developing Z_r [Gentine *et al.*, 2012].

Figure 4b shows the dependence of simulated Z_r on the time lag between energy (i.e., E_p) and water (i.e., P) availabilities within a year. Deeper roots are found in areas where the seasonal evolutions of energy and water availabilities are out-of-phase (e.g., the Mediterranean climate). In such areas, deeper roots allow a larger soil water storage capacity to hold infiltrated water during rainy season(s) to be used by plants during dry season(s). This phenomenon reflects one of the main strategies of plants to deal with short-term dry spells and is also a good indication of plant adaption to the climate [Ichii *et al.*, 2007; Ivanov *et al.*, 2012; Kleidon and Heimann, 1998].

To further highlight the importance of climate seasonality on Z_r , we take the evergreen broadleaf tropical forests over Amazonia as an example (Figure 5). A notable and interesting feature of the spatial Z_r distribution in Figure 2 is that a high spatial variability of Z_r in the Amazon basin, with the middle parts of the basin showing smaller Z_r values and the surrounding areas having much deeper roots. This spatial variability of Z_r is well related with the spatial heterogeneity of climate seasonality (Figure 5). Despite the fact

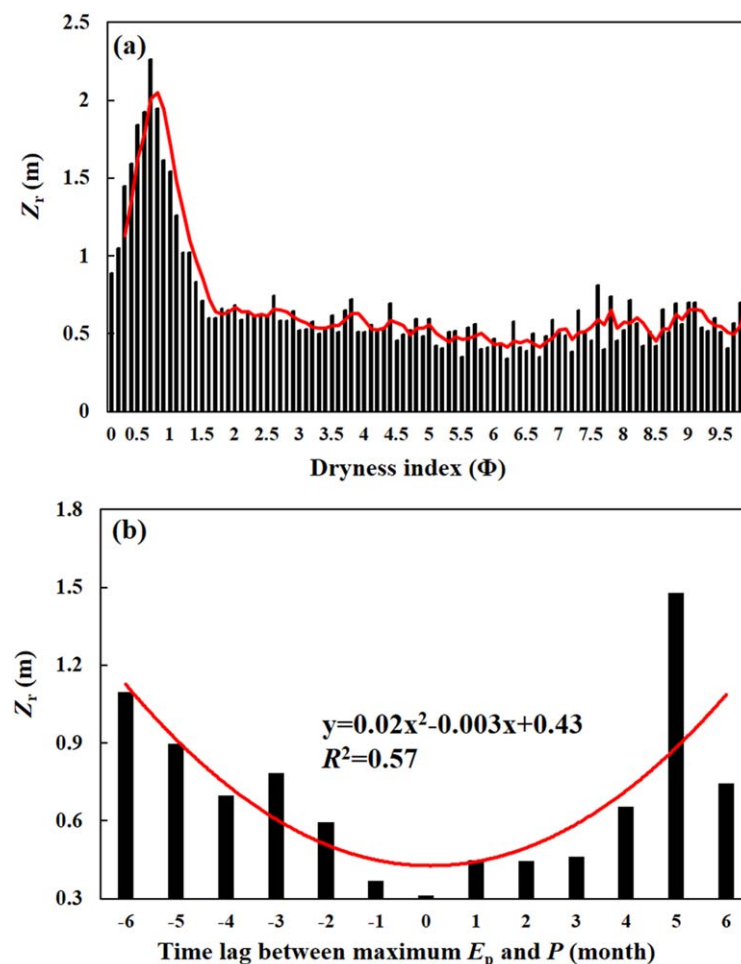


Figure 4. Relationship between simulated 0.5° resolution plant rooting depth and climatic factors across the globe. Effective rooting depth as a function of (a) dryness index and (b) time lag between the months of maximum precipitation and the maximum potential evapotranspiration. The red curve in Figure 4a represent the moving average of Z_r over 0.3 dryness index units along the dryness index gradient, and the red curve in Figure 4b is the best-fit quadratic function.

e.g., in western North America, central and middle-east Asia, and central Australia). Moreover, the shallow rooting depth in the middle and western parts of Amazonia is also captured in both $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ (although not so clear in $Z_{r,Schenk}$) (Figures 6a and 6c, and supporting information Figure S5).

The spatial pattern in the difference between Z_r estimates shows that the carbon cost-benefit model generally produces smaller Z_r values in comparison to $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ (Figures 6b and 6d). The largest differences are found in tropical and subtropical regions where savanna (e.g., the southeast Amazon, central Africa, and northern Australia) and cropland (e.g., the Indian Plain) are the major vegetation types. This difference can be better identified when comparing the three Z_r estimates at the biome level (Figure 7). It is observed that $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ share similar values for savanna and cropland and are much larger than $Z_{r,Guswa}$. For grasslands and deserts, $Z_{r,Schenk}$ exhibits larger values than the other two estimates that are of the similar magnitudes. For forest ecosystems, differences in Z_r estimates between the three studies are less than that for nonforest ecosystems, except for EBFs, where $Z_{r,Guswa}$ shows a substantially deeper rooting depth than the other two methods (Figure 7).

3.3. Hydrological Implications of Z_r

To test $Z_{r,Guswa}$ and to examine how different Z_r parameterizations affect hydrological model behavior (i.e., partitioning P between E and Q), we use the BCP model with different Z_r parameterizations to

that the entire region has the same vegetation type (i.e., evergreen broadleaf forest according to MOD12Q1 land classification), a similar mean climate condition (Figure 5a) and a nearly full vegetation coverage (Figure 5b), differences in climate seasonality have resulted in a highly diverse pattern of dry season length across the region (Figure 5c), which predominately controls the spatial variation of simulated Z_r in the area ($R^2 = 0.84$, Figures 5d and 5e).

3.2. Comparing Estimated Z_r With Other Studies

Two existing global Z_r estimates (i.e., $Z_{r,Kleidon}$ and $Z_{r,Schenk}$) are compared with Z_r estimated using the carbon cost-benefit model (referred to as $Z_{r,Guswa}$ herein). Overall, the three Z_r maps show a similar global spatial pattern: tropical and subtropical regions often have a deeper Z_r whereas Z_r in cold areas is much shallower (Figures 6a and 6c). For arid regions, $Z_{r,Kleidon}$ agrees reasonably well with $Z_{r,Guswa}$ but both estimates are much smaller than $Z_{r,Schenk}$ (typically 400 ~ 1200 mm less,

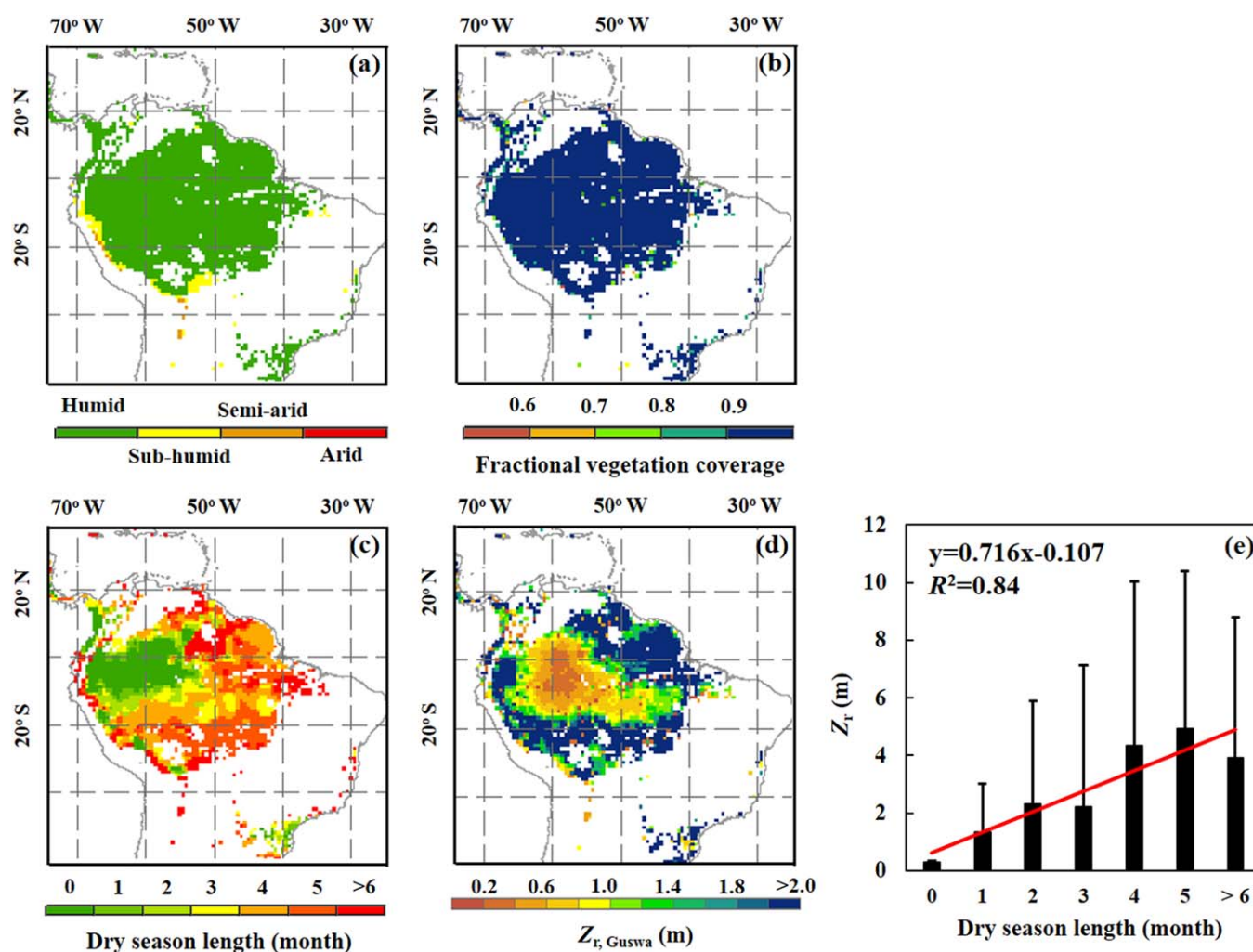


Figure 5. Influence of dry season length on Z_r in evergreen broadleaf forests within the Amazon basin. (a) Climate aridity zones, (b) fractional vegetation coverage [Zhu et al., 2016], and (c) dry season length for evergreen broadleaf forest in the Amazon basin. (d) Spatial pattern of Z_r in evergreen broadleaf forests within the Amazon basin, and (e) Z_r as a function of dry season length for the areas shown in Figures 5a–5c. Error bars and the red line in Figure 5e indicate one standard deviation and the best fit linear function, respectively. The dry season was identified when mean monthly potential evapotranspiration is larger than that of precipitation [Kleidon and Heimann, 1998].

estimate mean annual E globally. At the catchment scale, the estimated E from the BCP model combined with $Z_{r, Guswa}$ ($E_{BCP, Guswa}$) agrees very well with observed E from the catchment water balance (Figure 8a). On average, $E_{BCP, Guswa}$ could explain 94% of the spatial variations in the observed E , and the root-mean square difference (RMSD) and mean bias (defined as estimated mean minus observed mean) of $E_{BCP, Guswa}$ are 74 and 104 mm yr⁻¹, respectively (Figure 8a). This result suggests the applicability of the *Guswa's* carbon cost-benefit model of Z_r as estimated here, is effective for hydrological modeling across all global terrestrial ecosystems. By comparison, the performance of the BCP model is weaker when implemented with $Z_{r, Kleidon}$ and $Z_{r, Schenk}$ (Figures 8b and 8c), as can be seen in lower R^2 values (i.e., 0.85 for $E_{BCP, Kleidon}$ and 0.86 for $E_{BCP, Schenk}$), higher RMSDs (i.e., 132 mm yr⁻¹ for $E_{BCP, Kleidon}$ and 143 mm yr⁻¹ for $E_{BCP, Schenk}$) and mean biases (i.e., 168 mm yr⁻¹ for $E_{BCP, Kleidon}$ and 175 mm yr⁻¹ for $E_{BCP, Schenk}$).

Globally, at the 0.5° grid-box scale, $E_{BCP, Guswa}$ shows a consistent pattern with PML E , with R^2 , RMSD, and mean bias of 0.90, 125 mm yr⁻¹ and 27 mm yr⁻¹, respectively (Figure 10a). More than three quarters of the total land areas have an absolute difference lower than 100 mm yr⁻¹ (Figure 9c), further demonstrating the effectiveness of the $Z_{r, Guswa}$ in regulating the hydrological partitioning across the globe. In contrast, similar with the catchment results, the BCP model performance is worse when coupled with $Z_{r, Kleidon}$ and $Z_{r, Schenk}$ (Figures 10 and 11). Compared with the average

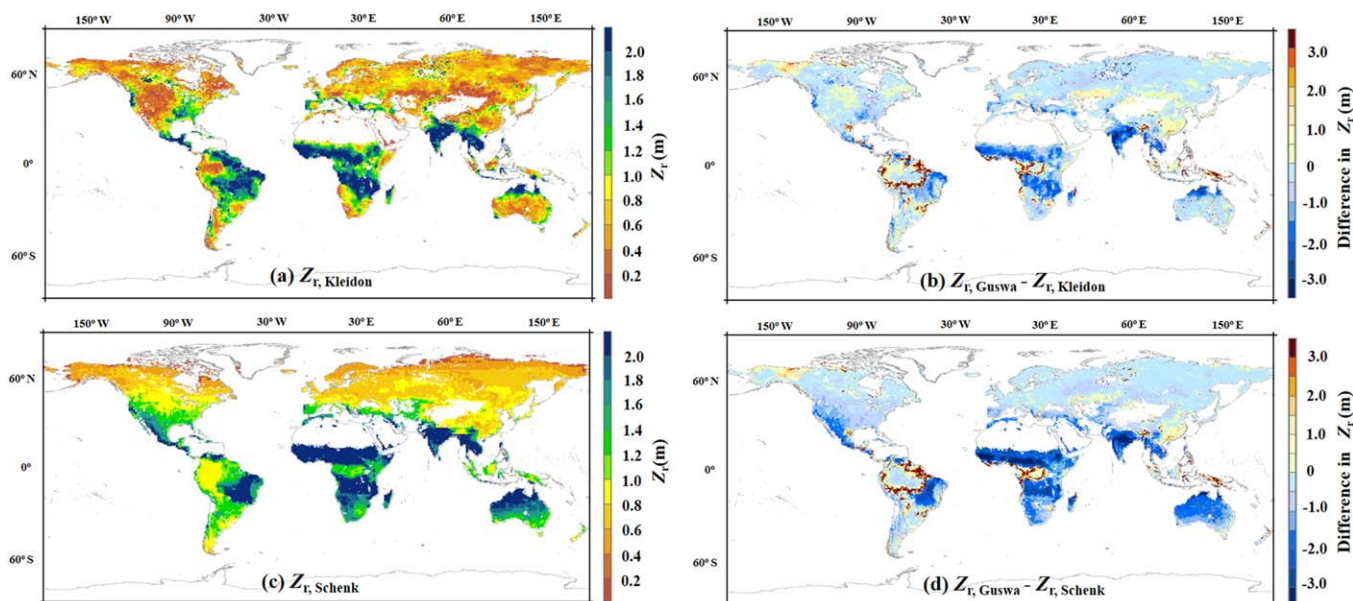


Figure 6. Comparison of Z_r estimates between studies. (a) Spatial distribution of Z_r estimated from an optimality model in Kleidon [2004] and (b) spatial pattern of the difference between Z_r , Guswa and that of Kleidon [2004]. (c) spatial distribution of Z_r estimated by upscaling in situ observations in Schenk and Jackson [2009] and (d) spatial pattern of the difference between Z_r , Guswa and that of Schenk and Jackson [2009].

of $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ used in the BCP model, implementation of $Z_{r,Guswa}$ reduces the RMSD of BCP E by 47 mm yr^{-1} (or 9.4%) and 50 mm yr^{-1} (or 10%), respectively (Figure 10).

The improvement of the BCP model performance through implementing $Z_{r,Guswa}$ as compared to $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ is consistent across biomes in terms of R^2 , RMSD, and mean bias using PML E as a reference, except for similar R^2 values among the three Z_r parameterizations in nonforest ecosystems (Figure 11). Nevertheless, despite similar values of R^2 , the RMSDs of $E_{BCP,Guswa}$ are substantially lower than that of $E_{BCP,Kleidon}$ and $E_{BCP,Schenk}$ in nonforest ecosystems, which can be mainly ascribed to a lower positive mean bias of $E_{BCP,Guswa}$ as a result of shallower $Z_{r,Guswa}$ in those ecosystems than the other two Z_r estimates (Figures 7 and 11).

4. Discussion

Z_r is an important parameter in hydrological and biogeochemical modeling to quantify the exchanges of carbon, water, and energy between the earth and the atmosphere. However, the determination of Z_r in current hydrological and biogeochemical models (especially when applied globally) is largely based on

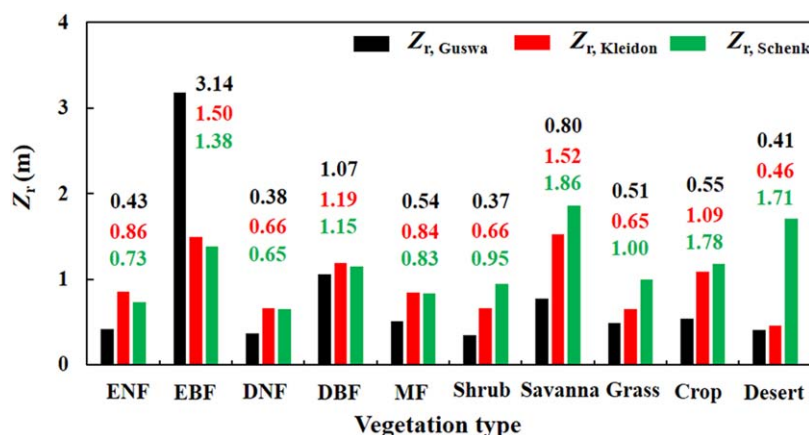


Figure 7. Comparison of biome-averaged rooting depth between studies, with values given in black, red and green for Z_r , Guswa, Z_r , Kleidon and Z_r , Schenk respectively. Vegetation-type abbreviations are given in section 2.4.

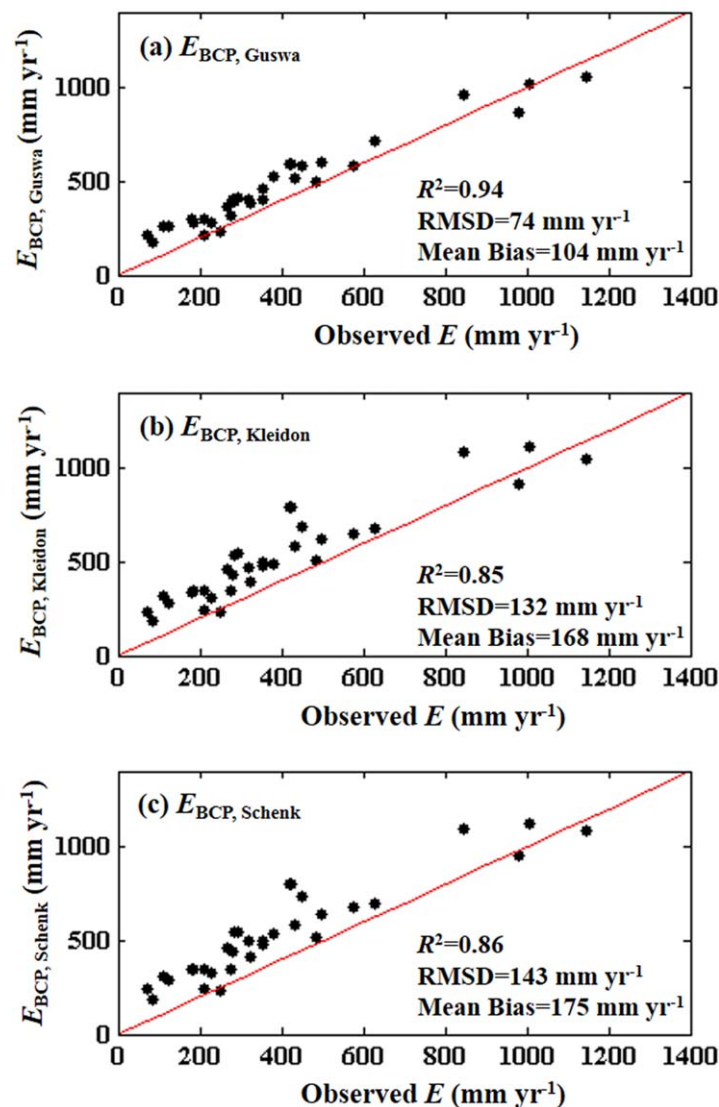


Figure 8. Validation of mean annual E estimates from the BCP model with different Z_r parameterizations across 32 global major catchments over 1984–2006. (a) E estimated from the BCP model with Z_r quantified in this study. (b) E estimated from the BCP model with Z_r of Kleidon [2004]. (c) E estimated from the BCP model with Z_r of Schenk and Jackson [2009]. The red lines are 1:1 line.

the look-up table approach, in which Z_r for each vegetation type is a fixed value and is quantified according to literature values of mean biome Z_r and soil texture information [Müller Schmied *et al.*, 2014; Sellers *et al.*, 1996; Wang-Erlandsson *et al.*, 2014; Zeng, 2001]. This approach of determining Z_r suffers from two main issues.

First, a fixed value of Z_r for each biome does not allow any variability of Z_r within each biome. By applying a carbon cost-benefit model, we clearly demonstrate that high Z_r spatial heterogeneity exists not only between biomes but also within each biome, which is strongly related with mean and seasonal climate variabilities (Figures 3–5). For the same soil texture, a deeper Z_r corresponds to a larger soil water storage capacity and thus provides the ecosystem with an enhanced potential water supply for vegetation use during the dry season. Our results highlight the importance of climate in shaping the spatial pattern of Z_r , which is consistent with previous findings [e.g., Collins and Bras, 2007; Guswa, 2008; Kleidon and Heimann, 1998; Ichii *et al.*, 2007]. In this regard, Z_r should preferably be classified on a water-deficit basis rather than on a biome basis [Kleidon and Heimann, 1998].

Second, the look-up table approach is problematic in the scale mismatch between Z_r observations that are used to obtain the biome averages and the real modeling. Observed Z_r usually represents Z_r of a single plant or several plants but does not represent Z_r over a modeling unit. Consequently, Z_r from look-up tables could differ greatly from the effective Z_r over a modeling unit that is more hydrologically and ecologically meaningful. The same problem also exists in $Z_{r, Schenk}$, which is an upscaled Z_r product based on observed rooting depth. This explains the overall larger values of $Z_{r, Schenk}$ than $Z_{r, Guswa}$ in particular for nonforest ecosystems where vegetation coverage is usually low (Figures 6 and 7). As for $Z_{r, Kleidon}$, although a similar modeling logic was adapted as in Guswa [2008] by incorporating a soil moisture dynamic model which allows its Z_r estimates reflect the dynamic balance between water supply and demand (e.g., the distinct Z_r pattern in the Amazon basin), it assumes a full vegetation cover and pursues the maximum potential vegetation productivity instead of actual productivity. Therefore, $Z_{r, Kleidon}$ may represent the potential rooting depth that is deeper than the actual rooting depth as estimated herein, especially in water-limited ecosystems (Figures 6 and 7).

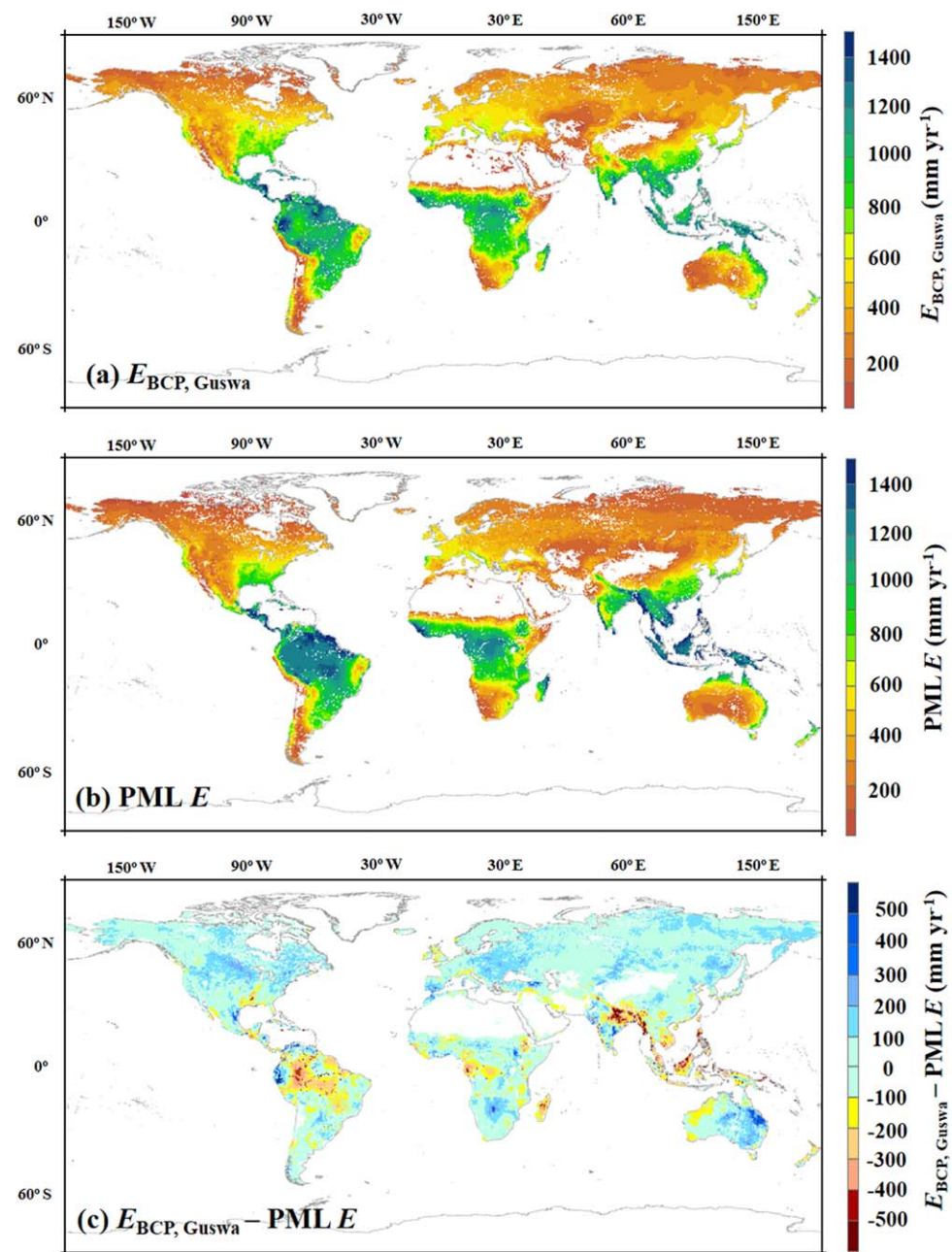


Figure 9. Comparison of mean annual $0.5^\circ E$ estimates from the BCP model with $Z_{r, \text{Guswa}}$ against E estimates from the PML model [Zhang *et al.*, 2016] over 1982–2010. (a) The spatial pattern of mean annual E estimates from the BCP model with $Z_{r, \text{Guswa}}$ over 1982–2010. (b) Illustrates the spatial pattern of mean annual PML E over 1982–2010. (c) The spatial pattern of the difference between mean annual $E_{\text{BCP, Guswa}}$ and PML E over 1982–2010.

The effective depth of plant roots has profound impacts on the hydrological cycle. For a given amount of rainfall, a deeper Z_r would result in a higher E and thus lower Q when all else are constant (e.g., the same soil properties). Here we use a highly transparent hydrological model to examine the effect of three Z_r parameterizations on the simulation of mean annual E . Our results show that, by incorporating $Z_{r, \text{Guswa}}$, the BCP model performs best of the three Z_r parameterizations at both the catchment and grid-box scales. This suggests the capability of *Guswa*'s model to estimate the hydrological, and presumably the ecological, effective rooting depth. In contrast, both $Z_{r, \text{Kleidon}}$ and $Z_{r, \text{Schenk}}$ have led to a significant overestimation of E by the BCP model, confirming our previous arguments that Z_r might be largely overestimated by *Kleidon* [2004] and *Schenk and Jackson* [2009].

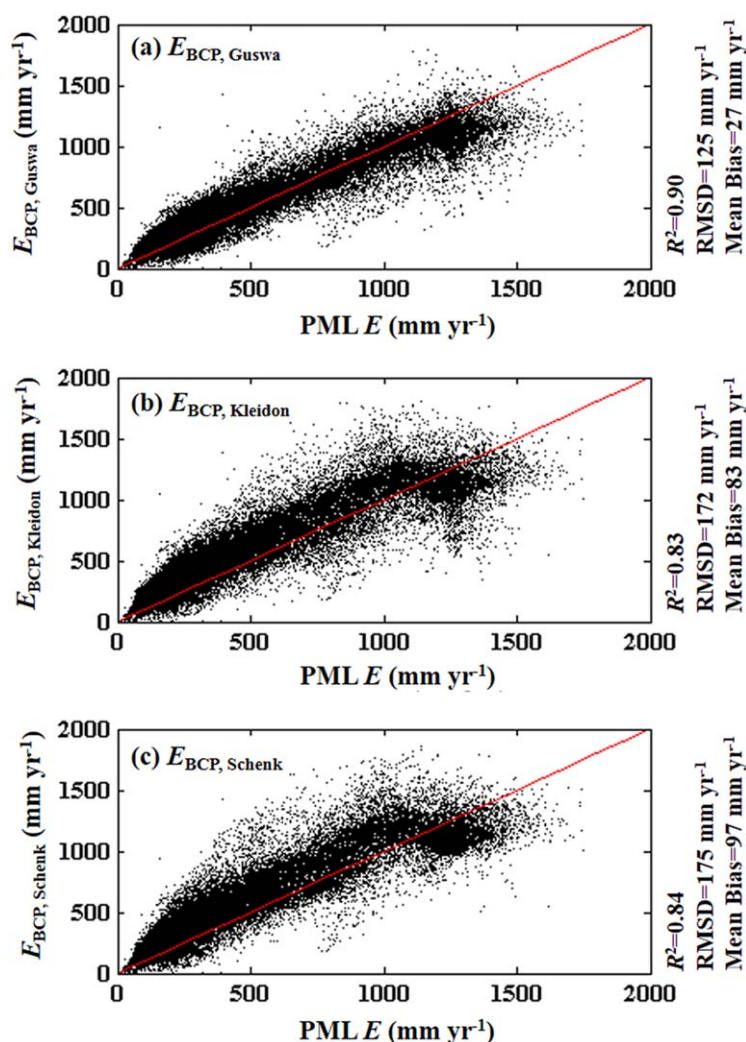


Figure 10. Comparison of mean annual E estimates from the BCP model with different Z_r parameterizations against E estimates from the PML model [Zhang et al., 2016] over 1982–2010. (a) E estimated from the BCP model with $Z_{r,Guswa}$ quantified in this study. (b) E estimated from the BCP model with Z_r of Kleidon [2004]. (c) E estimated from the BCP model with Z_r of Schenk and Jackson [2009]. Mean of PML E is 498 mm yr^{-1} . The red lines are the 1:1 lines.

estimated E by 62 mm yr^{-1} and 82 mm yr^{-1} (or 12.4% and 16.4%), in comparison to the use of $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ respectively, using Zhang et al.'s [2016] PML E as a benchmark.

Despite the overall good performance of $Z_{r,Guswa}$ in combination of the BCP model, to reproduce the global E pattern, the rooting depth estimated herein is inevitably subject to limitations. First, uncertainties in the input data/variable would introduce errors in our Z_r estimates. For example, the global WUE pattern was calculated based on ecosystem model estimates of GPP and E_T (their uncertainties have been discussed in many previous studies, e.g., Long et al. [2014] and Anav et al. [2015]), RLD and SRL were prescribed to their reported medium values and γ_r was taken from a biome-based look-up table. Nevertheless, sensitivity analysis suggests that the estimated Z_r is not very sensitive to uncertainties in WUE , RLD , SRL , and γ_r (Table 2). A 20% change in WUE , RLD , SRL , and γ_r similarly result in a typical 2% ~ 2.5% change in Z_r estimates. Higher sensitivities are found for θ and α , suggesting uncertainties in our Z_r estimates are primarily governed by those in the θ and α data sets. This could partly explain the unreasonable deep Z_r in northern Canada and eastern Alaska, where global forcing data sets often contain a higher uncertainties due to limited in situ observations to construct the data (e.g., see Figure 6 in Beck et al. [2016]). Second, ignoring the potential water supply from snow melt in cold regions would also lead to overestimation of Z_r (e.g., in northern Canada and eastern Alaska). Third, the Guswa's [2008] model as

The sensitivity of hydrological partitioning to variations in Z_r also differs greatly between climatic zones. The impact of catchment surface properties (including Z_r) on the long-term hydrological partitioning is mostly manifested in transitional climate zones (e.g., the 'equitant' climate region defined in McVicar et al. [2012], where Φ is between 0.76 and 1.35) but the impact of catchment surface properties diminishes to be less important than climate in extreme dry and wet areas. This is because in dry areas, water availability is generally the most limiting factor; whereas in wet zones, the surface hydrological partitioning is more regulated by energy availability [Budyko, 1974; Liang et al., 2015; McVicar et al., 2012; Roderick and Farquhar, 2011]. Our results clearly show that larger differences in simulated E between Z_r parameterizations are mostly concentrated in the middle E range, whereas differences in E are much smaller at the high and low E ends (Figures 8 and 10). If we only focus on the equitant climate zone, implementation of $Z_{r,Guswa}$ reduces the RMSD of the BCP

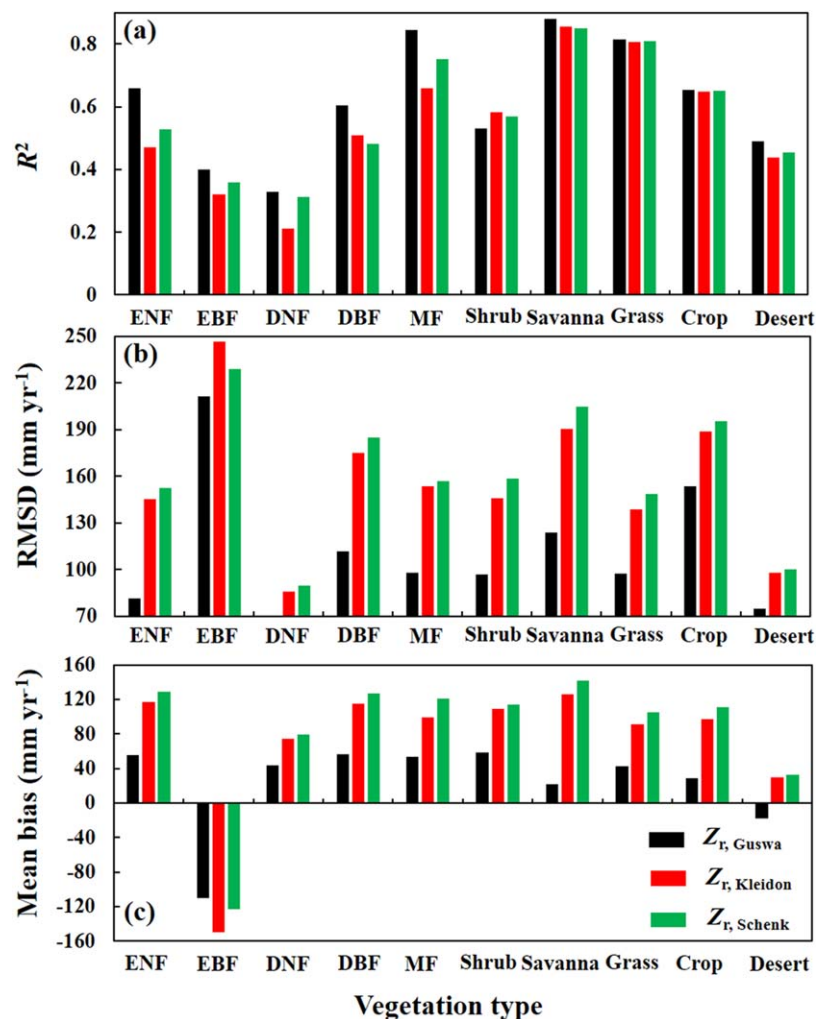


Figure 11. Validation of mean annual E estimates from the BCP model with different Z_r parameterizations over 1982–2010 for each biome, using Zhang et al.'s [2016] PML E as a reference. (a) R^2 , (b) root-mean-square difference and (c) mean bias, calculated as BCP-modeled E minus PML E .

used here implicitly assumes a uniform root water uptake ability along the root profile. Studies have shown that when soil moisture is limited, plants may respond to nonuniform stress conditions by increasing water uptake from relatively wet parts of the root zone [e.g., Jarvis, 1989; Skaggs et al., 2006]. Neglecting this compensation effect would lead us to overestimate the plant water stress and consequently overestimation of Z_r and E (Figure 8a). Fourth, our modeling approach focuses on the impact of balance between water supply and demand on the development of root system, whereas other factors, such as nutrient availability, are ignored. To account for nutrient limitation on root growth, a more sophisticated model is required to simultaneously consider both water and nutrient stresses. Nevertheless, if we assume that nutrients are taken up by plants as ions that are dissolved in the soil water, an explicit expression of the nutrient stress factor in our Z_r model becomes less important.

Table 2. Relative Sensitivity of Estimated Z_r to Input Variables ($\pm 20\%$ Changes) Across the Globe Over 1982–2010^a

Variable	WUE	RLD	SRL	γ_r	f_{GS}	θ	α
+20%	+2.0%	−2.0%	+2.0%	−2.0%	+2.0%	−13.0%	+14.6%
−20%	−2.5%	+2.5%	−2.5%	+2.5%	−2.5%	+13.4%	−15.0%

^aFor example, a 20% increase (decrease) in WUE results in a 2% increase (2.5% decrease) in Z_r estimates.

5. Conclusions

This study provides a global parameterization of the carbon cost-benefit model of Guswa [2008] to estimate the effective plant rooting depth across the globe. We find that despite a distinct Z_r between biomes, a high spatial heterogeneity of Z_r also exists within each biome, demonstrating the inadequacy of using look-up tables to define Z_r in the state-of-the-art land surface and hydrological models. Consistent with inverse modeling [Gentine et al., 2012], deeper rooting depths are found in areas where annual P is similar to E_p . For dry areas, Z_r is largely constrained by the availability of carbon and for extreme wet environments, shallow roots would effectively satisfy the atmospheric water demand. In addition to annual sums, the seasonality of water supply and demand also plays a critical role in determining Z_r , with areas having a longer dry season and water supply and demand are out-of-phase usually exhibiting a deeper Z_r .

We then adopt a parsimonious hydrological model (i.e., the modified BCP model) to test the effectiveness of $Z_{r,Guswa}$ in hydrological modeling. We find a generally good performance of the BCP model, when incorporated with $Z_{r,Guswa}$ in reproducing the observed and satellite-derived mean annual E across the globe (i.e., $R^2 = 0.94$, $RMSD = 74 \text{ mm yr}^{-1}$ for catchments, and $R^2 = 0.90$, $RMSD = 125 \text{ mm yr}^{-1}$ for grid-boxes). Compared with the two previous global Z_r estimates (i.e., $Z_{r,Kleidon}$ and $Z_{r,Schenk}$), implementation of $Z_{r,Guswa}$ in the BCP model significantly improves the model performance, especially over the equitant climate region where hydrological partitioning is most sensitive to variations in land surface properties (i.e., $RMSD$ reduces by 12.4% and 16.4% compared with $Z_{r,Kleidon}$ and $Z_{r,Schenk}$ respectively). Our results suggest that the new Z_r estimates (i.e., $Z_{r,Guswa}$) can be used in global hydrological modeling and potentially in biogeochemical models to simulate water, energy, and carbon fluxes over land.

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Erratum

This article has been updated to include a permanent link to the global rooting depth estimates for this paper. This version may be considered the official version of record.

In the originally published version of this manuscript, the Log10 terms in Equation 9, Equation 10, and Figure 1 were mistakenly typed in as Ln. The error has now been corrected, and this may be considered the official version of record.